

AI-Native Network Control for Remote Healthcare System Pipeline

1. Health Data Generation (Sender Node)

A healthcare traffic generator simulates a wearable monitoring device that continuously produces patient heart rate data.

- The sender generates **synthetic heart rate values**.
- Heart rate is categorized into three health states:
 - **NORMAL** (60–90 bpm)
 - **ALERT** (100–120 bpm)
 - **CRITICAL** (>120 bpm)
- Each packet contains:
 - sequence number
 - timestamp
 - heart rate value
 - health label

The packets are transmitted periodically using **UDP communication** to the receiver node.

2. Network Telemetry Collection

The receiver node continuously monitors the network performance.

For every time interval, the system calculates telemetry metrics such as:

- **Throughput**
- **Bandwidth usage**
- **Packet loss rate**
- **Delay jitter**
- **Queue length**

These measurements represent the **current condition of the network**.

The telemetry values are stored in CSV logs for analysis.

3. Feature Engineering

To capture temporal behaviour of the network, additional features are created using **rolling windows**.

Rolling averages are calculated for:

- packet loss rate
- jitter
- throughput

These features allow the model to observe **trends over time instead of single measurements**.

4. Machine Learning Based Network State Classification

A **Random Forest classifier** is trained using the collected telemetry dataset.

Input features include:

- bandwidth usage
- throughput
- packet loss rate
- jitter
- queue length
- rolling loss mean
- rolling jitter mean
- rolling throughput mean

The model predicts the network condition as one of the following states:

- **Stable**
- **Unstable**
- **Critical**

The trained model is stored as:

robust_network_model.pkl

5. Intelligent Network Decision Engine

At runtime, the receiver continuously performs the following steps:

1. Collect telemetry
2. Extract features
3. Predict network state using the trained ML model

The system then combines:

- **Predicted Network State**
- **Patient Health State**

to determine the appropriate communication strategy.

This creates **nine possible scenarios** based on the combination of network and health conditions.

6. Adaptive Communication Policy

Based on the predicted conditions, the receiver sends a **control command** to the sender.

Examples include

Network	Health	What To Send
Stable	Normal	Full ECG
Stable	Alert	Full ECG
Stable	Critical	Full ECG + priority
Unstable	Normal	Downsampled ECG
Unstable	Alert	Downsampled ECG
Unstable	Critical	Semantic alert + small ECG snippet
Critical	Normal	Semantic summary only
Critical	Alert	Semantic alert only
Critical	Critical	Immediate semantic critical alert

This enables **dynamic adaptation of communication behaviour**.

7. Sender Adaptation

Upon receiving the command, the sender modifies its transmission mode.

Possible transmission modes include:

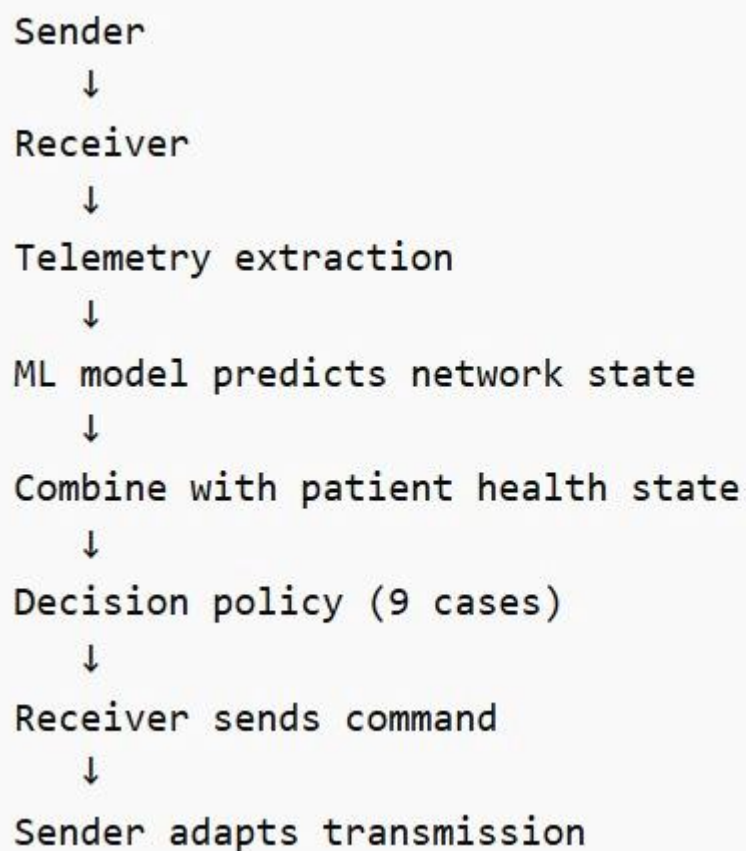
- **Full ECG Mode**
Continuous transmission of raw health data.

- **Priority ECG Mode**
Increased sampling rate for critical patients.
- **Downsampled ECG Mode**
Reduced data precision to lower network load.
- **Semantic Transmission Mode**
Only meaningful alerts or summaries are transmitted.

This reduces unnecessary network traffic while preserving critical information.

8. Closed-Loop AI-Native Network Control

The system forms a **closed-loop intelligent communication system**:



This feedback loop enables the network to **observe its own condition and adapt communication intelligently**.